

ARUS — Restore & Optimize HoursOfRestGrid

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CONTEXT

The Hours of Rest grid is the primary compliance data-entry screen. The previous prompt introduced auto-switching to mobile view and deprioritized the grid. REVERT those changes and restore the grid as the default, primary UI. Then apply the optimizations below to make the grid faster, clearer, and more usable without changing its fundamental layout or interaction model.

STEP 1 — REVERT MOBILE AUTO-SWITCH

In `client/src/components/HoursOfRestGrid.tsx`, remove this effect entirely if it was added:

```
// REMOVE THIS BLOCK:
React.useEffect(() => {
  const checkMobile = () => {
    if (window.innerWidth < 768 && viewMode === "month") {
      setViewMode("mobile");
    }
  };
  checkMobile();
  window.addEventListener("resize", checkMobile);
  return () => window.removeEventListener("resize", checkMobile);
}, []);
```

Ensure the default `viewMode` in `useHoursOfRestData` (or wherever initial state is set) is `"month"` — the full grid — not `"mobile"`.

The Month/Week/Mobile view toggle buttons should remain available so users can switch manually, but the grid must be the default on all screen sizes.

STEP 2 — PERFORMANCE OPTIMIZATIONS

2.1 — Extract grid template column string as a module-level constant

At the top of `HoursOfRestGrid.tsx`, outside the component function, add:

```
const HOUR_W = 24;
const GRID_COLS = `110px repeat(24, ${HOUR_W}px) 75px 75px` as const;
```

Replace every occurrence of the inline style:

```
style={{ display: "grid", gridTemplateColumns: `110px repeat(24, ${hourW}px) 75px
```

with:

```
style={{ display: "grid", gridTemplateColumns: GRID_COLS }}
```

Remove the local `const hourW = 24` declaration from inside the component.

2.2 — Memoize selected crew member lookup

Inside the component, after the hook destructuring, add:

```
const selectedCrewMember = React.useMemo(
  () => crew.find((c) => c.id === meta.crew_id),
  [crew, meta.crew_id]
);
```

Replace every inline `crew.find((c) => c.id === meta.crew_id)` call in JSX with `selectedCrewMember`.

2.3 — Memoize coverage and summary computations

Any derived values computed inline in JSX (coverage %, compliance stats) should be moved to `useMemo` calls above the return statement. Specifically, if there are any `.reduce()` or `.filter()` calls inside JSX, extract them:

```
const totalViolations = React.useMemo(
```

```
() => compliance.filter(c => !c.dayOK).length,  
[compliance]  
);
```

2.4 — Stabilize drag handlers with useCallback

Wrap `startDrag` and `onDrag` calls in the component (if they are defined inline rather than coming from the hook) in `React.useCallback`. If they come from the hook, ensure the hook returns stable references.

2.5 — Replace raw `<textarea>` in CSV mode with design system component

```
// Before:  
<textarea  
  className="w-full h-40 p-2 border rounded-md font-mono text-sm"  
  value={csv}  
  onChange={(e) => setCsv(e.target.value)}  
  data-testid="textarea-csv"  
>  
/  
// After:  
import { Textarea } from "@components/ui/textarea";  
<Textarea  
  className="h-40 font-mono text-sm"  
  value={csv}  
  onChange={(e) => setCsv(e.target.value)}  
  data-testid="textarea-csv"  
>  
/>
```

STEP 3 — GRID UX IMPROVEMENTS (preserving original layout)

3.1 — Add a sticky legend bar directly above the grid

This goes immediately before the grid scroll container (the div with `data-testid="rest-hours-grid"`), inside the `CardContent`:

```
<div className="flex flex-wrap items-center gap-x-5 gap-y-2 mb-3 px-1 text-xs tex  
  <span className="flex items-center gap-1.5 font-medium text-foreground">Key:</s
```

```

<span className="flex items-center gap-1.5">
  <span className="inline-block w-4 h-4 rounded bg-emerald-100 dark:bg-emerald-
    REST
  </span>
</span>
<span className="flex items-center gap-1.5">
  <span className="inline-block w-4 h-4 rounded bg-rose-100 dark:bg-rose-900 bo
    WORK
  </span>
</span>
<span className="flex items-center gap-1.5">
  <span className="inline-block w-4 h-4 rounded border-2 border-indigo-300 dark
    Night (20:00-06:00)
  </span>
</span>
<span className="flex items-center gap-1.5">
  <span className="inline-block w-4 h-1 rounded bg-rose-500" />
  Violation row
</span>
<span className="ml-auto italic">Click or drag to toggle cells</span>
</div>

```

3.2 — Add Clear All confirmation dialog

The Clear All button wipes an entire month silently. Wrap it in an AlertDialog:

```

import {
  AlertDialog, AlertDialogAction, AlertDialogCancel,
  AlertDialogContent, AlertDialogDescription, AlertDialogFooter,
  AlertDialogHeader, AlertDialogTitle, AlertDialogTrigger,
} from "@/components/ui/alert-dialog";

// Replace the existing Clear All button with:
<AlertDialog>
  <AlertDialogTrigger asChild>
    <Button
      variant="outline"
      size="sm"
      className="border-slate-300 text-slate-700 hover:bg-slate-50 dark:text-slat
      data-testid="button-clear-all"
      title="Clear all hours in the grid"
    >
      Clear All
    </Button>
  </AlertDialogTrigger>
  <AlertDialogContent>
    <AlertDialogHeader>
      <AlertDialogTitle>Clear all rest hours?</AlertDialogTitle>

```

```

<AlertDialogDescription>
  This will clear all rest hours for{" "}
  <strong>{selectedCrewMember?.name ?? "this crew member"}</strong> in{" "}
  <strong>{meta.month} {meta.year}</strong>.
  <br />
  You can immediately undo this with the Undo button.
</AlertDialogDescription>
</AlertDialogHeader>
<AlertDialogFooter>
  <AlertDialogCancel>Cancel</AlertDialogCancel>
  <AlertDialogAction
    onClick={clearAll}
    className="bg-destructive hover:bg-destructive/90"
  >
    Clear All Hours
  </AlertDialogAction>
</AlertDialogFooter>
</AlertDialogContent>
</AlertDialog>

```

3.3 — Make Save to Database the visually dominant action

Change the Save to Database button from outline styling to a filled primary button:

```

// Find the Save to Database button and change:
// variant="outline" or any outline class → variant="default"
// Replace its className with:
className="bg-blue-600 hover:bg-blue-700 text-white shadow-md hover:shadow-lg tra
// Keep data-testid="button-upload-grid" and title attribute unchanged

```

3.4 — Undo/Redo buttons: add step count tooltips

```

import {
  Tooltip, TooltipContent, TooltipProvider, TooltipTrigger,
} from "@components/ui/tooltip";

// Replace the undo button with:
<TooltipProvider>
  <Tooltip>
    <TooltipTrigger asChild>
      <Button
        variant="outline"

```

```

        size="sm"
        onClick={undo}
        disabled={historyIndex <= 0}
        title="Undo"
        data-testid="button-undo"
      >
        <Undo className="w-4 h-4" />
      </Button>
    </TooltipTrigger>
    <TooltipContent side="bottom">
      {historyIndex <= 0
        ? "Nothing to undo"
        : `Undo (${historyIndex} step${historyIndex !== 1 ? "s" : ""} available)`
      }
    </TooltipContent>
  </Tooltip>
</TooltipProvider>

// Replace the redo button with:
<TooltipProvider>
  <Tooltip>
    <TooltipTrigger asChild>
      <Button
        variant="outline"
        size="sm"
        onClick={redo}
        disabled={historyIndex >= history.length - 1}
        title="Redo"
        data-testid="button-redo"
      >
        <Redo className="w-4 h-4" />
      </Button>
    </TooltipTrigger>
    <TooltipContent side="bottom">
      {historyIndex >= history.length - 1
        ? "Nothing to redo"
        : `Redo (${history.length - 1 - historyIndex} step${history.length - 1 -
      }
    </TooltipContent>
  </Tooltip>
</TooltipProvider>

```

3.5 — Add beforeunload guard for unsaved changes

Inside the component, add:

```

React.useEffect(() => {

```

```
const handler = (e: BeforeUnloadEvent) => {
  if (saveStatus === "unsaved") {
    e.preventDefault();
    e.returnValue = "";
  }
};
window.addEventListener("beforeunload", handler);
return () => window.removeEventListener("beforeunload", handler);
}, [saveStatus]);
```

3.6 — Save success toast notification

At the top of the component add:

```
import { useToast } from "@/hooks/use-toast";
const { toast } = useToast();
```

Find where `saveStatus` transitions to `"saved"` (likely inside `useHoursOfRestData` or in a mutation `onSuccess` callback). If the save is triggered by the `upload` function exposed by the hook, wrap the button's `onClick` to show a toast on success.

If the hook exposes an `onSaveSuccess` callback prop or similar, use that. Otherwise, add a `useEffect` watching `saveStatus`:

```
const prevSaveStatus = React.useRef(saveStatus);
React.useEffect(() => {
  if (prevSaveStatus.current === "saving" && saveStatus === "saved") {
    toast({
      title: "Rest hours saved",
      description: `Saved for ${selectedCrewMember?.name ?? "crew member"} — ${me}`,
    });
  }
  prevSaveStatus.current = saveStatus;
}, [saveStatus]);
```

3.7 — Replace checkmark emoji with icon

```
// Before:
<span className="text-lg">✓</span>
// After:
import { CheckCircle2 } from "lucide-react";
```

```
<CheckCircle2 className="h-4 w-4 text-emerald-600 dark:text-emerald-400 inline-bl
```

3.8 — Violation tooltip: show which rule was violated

Each row has a compliance indicator bar at the bottom. Add a `title` attribute to this bar div so hovering reveals which rule failed:

```
// Find the compliance bar div at the bottom of each row:
<div
  className={`h-1 transition-all duration-300 ${dayOK ? "bg-gradient-to-r from-em
  style={{ marginBottom: 2 }}
  // Add this:
  title={dayOK ? "Compliant" : (() => {
    const reasons: string[] = [];
    if (c.restTotal < 10) reasons.push(`Rest ${c.restTotal}h < 10h minimum`);
    if (c.minRest24 < 10) reasons.push(`24h window: ${c.minRest24.toFixed(1)}h <
    if (!c.splitOK) reasons.push("Rest blocks: split rule violated");
    return reasons.join(" · ") || "Violation";
  })()}
>
```

Also add the same title to the date cell that has the red left border:

```
// On the date cell div, add:
title={!dayOK && liveCheck ? (() => {
  const reasons: string[] = [];
  if (c.restTotal < 10) reasons.push(`Only ${c.restTotal}h rest`);
  if (c.minRest24 < 10) reasons.push(`24h window: ${c.minRest24.toFixed(1)}h`);
  if (!c.splitOK) reasons.push("Block split violated");
  return "⚠ " + (reasons.join(" · ") || "Violation");
})() : undefined}
```

3.9 — Add keyboard navigation to grid cells

On every grid cell div, add keyboard support so users can toggle cells without a mouse:

```
// Add these props to each cell:
tabIndex={0}
role="checkbox"
aria-checked={isRest}
aria-label={` ${isRest ? "Rest" : "Work"} at ${String(h).padStart(2, "0")}:00${isN
```



```

onKeyDown={ (e) => {
  if (e.key === " " || e.key === "Enter") {
    e.preventDefault();
    startDrag(actualIndex, h);
  }
}}

```

This adds accessibility without changing the visual layout at all.

3.10 — Selected day highlight: show day stats in a fixed info bar

When a user clicks a row (`setSelectedDay`), currently nothing visible happens except a blue background tint on the row. Add a compact stats bar below the grid that shows the selected day's numbers:

Add this state at the top of the component:

```

// selectedDay already exists from the hook — use it

```

Add this element immediately AFTER the grid scroll container div (after the closing div of `data-testid="rest-hours-grid"`):

```

{selectedDay !== null && compliance[selectedDay] && (
  <div className="mt-3 p-3 rounded-lg border bg-blue-50 dark:bg-blue-950/30 borde
    <span className="font-semibold text-blue-800 dark:text-blue-200">
      {displayRows[selectedDay]?.date}
    </span>
    <span className={compliance[selectedDay].restTotal >= 10
      ? "text-emerald-700 dark:text-emerald-400"
      : "text-rose-700 dark:text-rose-400"}>
      Rest: <strong>{compliance[selectedDay].restTotal}h</strong>
      {compliance[selectedDay].restTotal < 10 && " ⚠ (min 10h)"}
    </span>
    <span className={compliance[selectedDay].minRest24 >= 10
      ? "text-emerald-700 dark:text-emerald-400"
      : "text-rose-700 dark:text-rose-400"}>
      24h window: <strong>{compliance[selectedDay].minRest24.toFixed(1)}h</strong>
    </span>
    <span className={compliance[selectedDay].splitOK
      ? "text-emerald-700 dark:text-emerald-400"
      : "text-rose-700 dark:text-rose-400"}>
      Block split: <strong>{compliance[selectedDay].splitOK ? "OK" : "Violated"}<
    </span>
    <button

```

```

        onClick={() => setSelectedDay(null)}
        className="ml-auto text-xs text-muted-foreground hover:text-foreground"
        aria-label="Dismiss day detail"
      >
        ✕
      </button>
    </div>
  )}

```

3.11 — Week view: fix off-by-one crash on last partial week

Fix the Next Week button's upper bound:

```

// Before:
onClick={() => setWeekOffset(Math.min(Math.floor(rows.length / 7), weekOffset + 1)}
// After:
onClick={() => setWeekOffset(Math.min(Math.ceil(rows.length / 7) - 1, weekOffset

```

Also guard the row render against undefined rows:

```

// Before:
{displayRows.map((r, ri) => {
// After:
{displayRows.filter(Boolean).map((r, ri) => {

```

3.12 — Deprecation notice on `client/src/components/hours-of-rest/utils.ts`

Add a JSDoc deprecation comment at the top of the file:

```

/**
 * @deprecated
 * Import directly from `@shared/lib/rest-grid-utils` instead.
 * This barrel exists for backward compatibility only and will be removed.
 * TODO: Migrate all consumers and delete this file.
 */

```

STEP 4 — LEAVE UNCHANGED

Do NOT change any of the following:

- The grid's visual layout, column widths, or cell dimensions
 - The hour column count (24 columns)
 - The date column width (110px)
 - The Rest/Work totals columns (75px each)
 - The sticky header structure
 - The compliance bar height or position
 - The month/week/mobile toggle buttons (keep all three, just don't auto-switch)
 - The drag-to-paint mouse interaction model
 - The `useHoursOfRestData` hook internals
 - Any file in `@shared/`
 - Any API endpoints or query keys
 - Any test files
 - The `HoursOfRest.tsx` simple calendar component (separate file, leave it alone)
-

VALIDATION CHECKLIST

After applying all fixes, verify:

- Default view on load is the full month grid (not mobile view)
- Grid layout and dimensions are unchanged from original
- Month/Week/Mobile toggle still works manually
- `GRID_COLS` constant defined at module level, no inline template string on rows
- `selectedCrewMember useMemo` replaces all inline `crew.find()` calls
- Legend bar appears above the grid
- Clear All shows `AlertDialog` confirmation before clearing
- Save button is filled blue, visually distinct from utility buttons
- Undo tooltip shows step count (e.g. "Undo (3 steps available)")
- Redo tooltip shows step count

- Navigating away with unsaved changes shows browser warning
- Successful save shows toast: "Rest hours saved for [name] — [month year]"
- Checkmark emoji replaced with CheckCircle2 icon
- Hovering the violation bar shows which rule(s) failed
- Clicking a row shows the day stats bar below the grid
- Clicking × on the day stats bar dismisses it
- Week view does not crash on last partial week of 31-day month
- Grid cells respond to Space/Enter keyboard toggle
- raw textarea in CSV mode replaced with Textarea component
- utils.ts has @deprecated JSDoc comment